

When Online Attention Spills Over: Consumer Goods, Platform Control, and Government Response in China

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Abstract—In order to examine how social media originated controversies in China move beyond platform in term of specific discussion and become associated with offline spillovers in consumer goods, markets, and government response. Rather than treating “online trends” as a broad or self-evident category, we will focus on high visibility controversies that begin or intensify in Chinese social media environment and then become visible across a wider public attention space, including news coverage, search behavior, investor attention, consumer behavior, and official response.

We developed a threshold–pathway–spillover framework. The claim is that online controversies in China do not become consequential simply because they are popular on social media. They become consequential when they break through a governed visibility environment and become legible to consumers, markets, or state actors. The first three cases focus on consumer goods spillovers: the 2024 Nongfu Spring controversy as a brand-boycott and reputation damage pathway, the 2023 Japan wastewater salt panic as a panic-buying and supply reassurance pathway, and the late 2022 Lianhua Qingwen episode as a demand of medicine and pharmaceutical-repricing pathway. The fourth case which is the Urumqi fire and Zero COVID reversal is treated separately as a boundary case, aim to show how politically sensitive attention can also break through platform control and become associated with government response.

Because high-quality historical data from Weibo, Douyin or WeChat, and Xiaohongshu are difficult to obtain, this paper does not claim to directly measure domestic Chinese social media visibility. Google Trends is used only as a secondary proxy for broader cross platform public salience after a controversy has already moved beyond the original platform environment. The analysis put together Google Trends with event timelines, media reports, market data, policy actions, and survey evidence from 104 respondents with experience in Chinese social media and some familiarity with U.S. or Western social media. The paper does not make strong causal claims, instead, it identifies patterned associations between visibility breakthroughs and offline spillovers under conditions of partial platform data.

I. INTRODUCTION

A. Background

The controversies in social media do not always remain inside. In many cases, a dispute that begins as scattered online discussion becomes visible across platforms, news coverage, investor attention, consumer behavior, and official response. Where we aim to studies that process in China, where platform visibility is shaped not only solely by user activity, but also by recommendation systems, trending lists, moderation practices,

search filtering, and state regulation. The central issue is therefore not simply whether a topic appears online. The critical question is that when an online controversy becomes visible enough to matter outside the platform.

This distinction is especially important in the Chinese context. Public controversies may circulates through Weibo, Douyin, WeChat, Xiaohongshu, short video re-posting, screenshots, private group chats, and news coverage at the same time. Yet these channels are not equally observable to us. A historical platform data with high quality from Chinese social media platforms are difficult to collect consistently, especially after an event has passed. For example, the case Urumqi Fired is under censorship. For that reason, this research does not attempt to reconstruct the full domestic platform ecology of each controversy. What we trying to do is to study how selected controversies moved from platform specific discussion into broader public salience and whether that broader visibility was temporally associated with consumer goods or government spillovers.

By doing such, it also avoids a common measurement problem. Google Trends cannot be used as a direct substitute for Weibo, Douyin, WeChat, or Xiaohongshu data, especially because Google is not normally accessible to ordinary users in mainland China and Google Trends measures normalized search interest rather than social-media discussion volume [1]–[3]. In this paper, Google Trends will be used more narrowly: as a secondary indicator of externalized, cross platform attention after a China related controversy has become visible beyond its original domestic social media environment. The data in Google Trends are interpreted alongside media reports, market indicators, policy timelines, and survey responses, not as a stand-alone measure of Chinese public opinion.

B. Research Question

In China, how do social media originated controversies move beyond platform specific discussion into broader public attention, and how are these visibility breakthroughs associated with consumer goods spillovers and government response?

We focuses mainly on consumer goods and market spillovers. The first three cases examine how online visibility is associated with brand damage, panic buying, and product-

linked demand. A fourth case, which is the Urumqi fire and Zero COVID reversal, will be a boundary case. Because this case is not relevant to consumer goods. It is included because it shows how politically sensitive attention can also exceed platform control and become associated with policy response.

C. Argument

What we argue is that in China online controversies do not influence offline behavior because they are trending on social media. They become consequential because when they break through platform level visibility or censorship controls and spread into broader public attention. Once that visibility threshold is crossed, then such event will be form of spillover depends on the type of event. Brand controversies may produce reputation damage and market decline. The fears in public health may produce panic buying and supply reassurance. Medicine related attention may produce demand expectations and stock repricing. Last but not least, politically sensitive crises may generate policy pressure and government response.

The contribution of this paper is not simply a claim that social media directly causes every observed market or policy outcome. The more cautious claim is that visibility breakthroughs are temporally associated with different forms of offline spillover. The analysis identifies these associations through a structured comparison of cases, not through a full causal model.

D. Contributions

The study makes four contributions. First, the study narrows the object of analysis. Instead of treating “online trends” as a broad category, we focus on social media originated controversies that will move beyond platform specific discussion, then becomes visible in a wider public attention environment. By such it separates ordinary online discussion from high visibility events that are more likely to generate observable spillovers.

Second, the study clarify measurement role of Google Trends. Google Trends is not use as a direct measure of domestic Chinese social media activity. In contrast, Google Trends is treated as a secondary indicator of broader cross platform salience after an event that is already moved beyond its original platform. The use of Google Trend measure is interpreted together with market data, event timelines, policy actions, media reports, and survey responses.

Third, what this study distinguishes among different spillover pathways is that not treating offline impact as a single outcome. The Nongfu Spring case illustrates a brand boycott and reputation damage pathway; the salt panic case illustrates a panic buying and supply reassurance pathway; and the Lianhua Qingwen case illustrates a medicine demand and pharmaceutical repricing pathway. The Urumqi fire is included separately as a boundary case, showing how politically sensitive visibility can become associated with government response rather than consumer goods behavior.

Lastly, the study adds survey. The survey is not nationally representative and is not used as causal evidence. Its purpose is

more niche: it helps assess whether the selected events were recognizable among respondents with Chinese social media experience and whether respondents perceived algorithms, trending lists, and government response as shaping event visibility. They survey will strengthens the study by use of partial platform data without overstating what the survey can prove.

II. RELATED LITERATURE

A. Platform Governance and Visibility Control in China

The study is builds on recent scholarship that treats platform visibility as a governed process rather than a purely organic outcome of user discussion. Online visibility is shaped by recommendation systems, hot-search lists, search filtering, content moderation, platform responsibility, and state regulation in China. The 2022 Internet Information Service Algorithmic Recommendation Management Provisions are especially relevant because they each covers personalized recommendation, ranking and selection, search filtering, dispatching and decision-making, and there are also other algorithmic technologies used to distribute information to users [4]. The provisions also restrict the manipulation of topic lists, hot-search terms, search-result rankings, and other forms of information presentation that could influence online public opinion [4]. This supports our claim that attention in China does not circulate in a neutral information market. It moves through a visibility environment where amplification and restriction are both governed.

There is also a recent scholarship on China’s algorithmic regulation makes this point more explicit. Yang and Yao argue that the social media in China’s algorithmic recommendation rules extend state oversight into the recommendation systems used by major online platforms, making algorithmic distribution a formal regulatory object rather than a purely technical design choice [5]. Xu and Yu’s work on internet and digital media governance in the Xi Jinping era provides a broader institutional background, emphasizing the centrality of state authority, platform responsibility, and regulatory control in China’s digital media system [6]. Chin, Park, and Li’s comparative study of false-information governance in Chinese and American social media platforms also helps clarify that platform governance differs across institutional settings, especially in the balance between state regulation, platform self-regulation, and co-regulation [7]. Together, these studies support the theoretical basis for ours governed-visibility framework: online controversies in China may spread quickly, but their spread is mediated by platform infrastructures and regulatory constraints.

These researches also help clarify the meaning of platform control in this study. Platform control does not mean that all sensitive information disappears, also does not mean that all viral attention is artificially manufactured. Rather, control refers to a set of institutional and technical mechanisms that would shape whether a topic becomes visible, how long it remains visible, and which public can easily encounter it. Even though algorithms and trending lists may amplify a

topic, meanwhile moderation and search filtering may restrict or redirect it. This is a dual process that is central to the study's threshold argument: an online controversy becomes consequential when it breaks through this controlled visibility environment and becomes legible to consumers, markets, or government.

B. Online Firestorms and Thresholds of Public Attention

The research on online firestorms provides the closest conceptual foundation for our threshold logic. Kim et al.'s comparative study of Weibo and Twitter shows that online firestorms differ across national and platform contexts, including differences in targets, expressive styles, and the forms of public reaction that become visible [8]. Kim's findings are important for this study because they suggest that an online controversy's visibility is not solely determined by user anger. Those visibilities are also shaped by platform environment, topic category, and broader political and media conditions.

This study widens the insight by asking what happens after a controversy crosses certain visibility threshold. Much of the online-firestorm literature focuses on the online episode itself: how public criticism forms, spreads, and intensifies. The present studies instead follow the controversy beyond the platform and ask whether high visibility attention becomes associated with offline spillovers in consumer behavior, market response, or government action. In conclusion, our study does not treat virtuality as the endpoint of analysis. It treats visibility breakthrough as an intermediate stage between platform discussion and offline consequence.

In Chinese context, the threshold concept is very useful because online criticism could exist without becoming broadly visible or politically consequential. Many controversies may circulate inside a platform, within screenshots, or among private networks without creating measurable spillover. In contrast, a controversy that moves across social media, news coverage, search behavior, investor attention, and official response becomes more legible as a public event. That's the reason why our threshold-pathway-spillover framework is designed to capture that difference.

C. Consumer Nationalism, Brand Boycotts, and Consumer Video Activism

In this paper, the consumer-goods cases also draw on recent work about consumer nationalism and digital consumer activism in China. Liao and Xia's study of the anti-Lotte boycott shows how Chinese internet users used digital media to coordinate a nationalist consumer boycott and connect purchasing behavior with national identity [9]. Their analysis is directly relevant to the Nongfu Spring case because it shows that brand controversies in China may become more than ordinary consumer disputes. They can become public performances of identity, loyalty, and national belonging.

Research on consumer video activism further supports this paper's claim that consumer controversies can move from individual dissatisfaction to broader public visibility. Yu's

empirical study of consumer video activism shows how Chinese consumers use short-video platforms to protest against businesses, attract attention, and seek responses from media or government departments [10]. Yu and Xu's later work on the emergence of a Chinese consumer sphere similarly argues that short-video-based social media platforms have become tools for consumers to protect their rights and make commercial grievances publicly visible [11]. These studies help explain why consumer-goods controversies can spill beyond ordinary purchasing choices. When a brand dispute becomes visible through social media, it may generate reputational pressure, consumer avoidance, or market reassessment.

This research supports our interpretation of the Nongfu Spring controversy as a brand boycott and reputational-damage pathway. The case is not treated as proof that online attention alone caused the firm's stock movement. Instead, it is interpreted as a case where social media originated criticism, brand identity, media attention, and investor attention became temporally aligned. The consumer nationalism and consumer activism literature makes this pathway plausible by showing that Chinese digital consumers can use online visibility to pressure firms and attach broader social meanings to consumption.

D. Online Attention, Panic Buying, and Market Response

Another recent literature links online attention to consumer behavior and market response. Studies of Chinese financial markets show that stock returns, trading volume, and commodity market reactions could be associated with social media sentiment, investor attention, and media sentiment. Liu et al. examine China's commodity futures market and find that social-network sentiment predicts commodity futures returns, while investor attention has a significant positive relationship with returns and absolute returns [12]. This supports our treatment of the salt panic as a case where online fear and market attention moved together around a consumer goods shock.

There are also other studies that provide broader support for the market spillover mechanism. Wang et al. find that social media sentiment can affect same-day stock returns by using experimental evidence from an online message forum, although the effect is short-lived [13]. Duan, Liu, and Wang study COVID-19 sentiment in China using official news media and Sina Weibo data, finding that COVID-related sentiment indices predicted Chinese stock returns and turnover rates during the early pandemic period [14]. These studies do not prove causality in our cases, but they support the more cautious claim that online attention and sentiment can become market relevant information.

Research on panic buying and public-health fear is also relevant to the salt panic and Lianhua Qingwen cases. Li et al.'s study of Chinese public panic buying at the beginning of the COVID-19 outbreak shows that perceived risk, social-media use, and connection with close others contributed to panic-buying behavior [15]. Hou et al.'s social-media surveillance preprint uses Sina Weibo, Baidu search, and Ali e-commerce marketplace data to track public attention, risk perception,

emotional response, protective behavior, and panic buying during the COVID-19 outbreak [16]. Because Hou et al. is a preprint, this paper uses it only as a methodological example rather than as core peer-reviewed evidence. Its value is to show what an ideal China based data design might look like: combining native social-media data, domestic search data, and e-commerce indicators. Because such historical platform data are not consistently available for all cases in this project, the present paper instead uses Google Trends, event timelines, market data, policy actions, and survey responses as combined evidence.

Together, these studies support our interpretation of the salt panic and Lianhua Qingwen cases. In the salt panic, online fear became associated with panic buying, salt-stock volatility, and government supply reassurance. In the Lianhua Qingwen case, reopening COVID era health uncertainty became associated with product specific attention, medicine demand expectations, pharmaceutical repricing, and later correction. The literature suggests that these patterns are plausible forms of online to offline spillover.

E. Government Response and the Boundary Policy Case

The reason why Urumqi fire is treated as a boundary case is because its main spillover not consumer purchasing behavior but government response. Recent work on protest and information control in China helps explain why politically sensitive visibility matters even in a controlled platform environment. Zheng and Meng's work on responsiveness and social protest in China shows that state response to public pressure is selective and institutionally shaped, rather than automatic [17]. This supports the paper's cautious interpretation of the Urumqi case: the event did not single-handedly cause the Zero COVID reversal, but it became part of a broader pressure environment in which public visibility was politically significant.

Information control research also clarifies the restrictive side of the platform environment. Ruan, Knockel, and Crete-Nishihata show that Chinese information control operates not only through censorship but also through public punishment and signaling repression, especially when online expression challenges government credibility or authority [18]. This helps explain why the Urumqi case should be treated separately from the consumer-goods cases. Its pathway involves politically sensitive attention, collective visibility, and state response, not ordinary market demand.

This literature supports the boundary case logic. The Urumqi case does not show that online attention directly caused policy reversal. Instead, it shows that even in a highly controlled platform environment, some politically sensitive controversies may become visible enough to enter the government's response environment. This strengthens the paper's broader claim that platform control does not eliminate online spillover. It changes the threshold and pathway through which spillover occurs.

III. CASE SELECTION

The reason we select these four cases because we want these serve as sample of recent China related controversies in the post pandemic period. These cases were selected because each case satisfies four conditions: first, a clear triggering event. Second, visible discussion on or around Chinese socialmedia platforms. Third, observable spillover into consumer behavior, market indicators, or government response, and last enough public information to reconstruct a basic event timeline.

The first three were selected to serve core consumer goods comparison. What the Nongfu Spring controversy captures is a brand boycott and reputational damage pathway. Meanwhile, Japan wastewater salt panic captures a panic-buying pathway in which online fear became associated with consumer hoarding, salt-stock volatility, and government supply reassurance. Lastly, Lianhua Qingwen episode captures a medicine demand pathway in which reopening COVID era health concerns became associated with product specific attention and pharmaceutical-stock repricing.

The Urumqi fire will be included as a boundary case. Because Urumqi fire main spillover was not consumer purchasing behavior, but policy pressure and government response. It is analytically useful because it shows the outer limit of the paper's argument: when an online controversy becomes politically sensitive and highly visible, platform control may shape its visibility, but it may not fully prevent public pressure from becoming legible to the state.

IV. MEASUREMENT STRATEGY AND GOOGLE TRENDS LIMITATION

A central measurement challenge in this project is the limited availability of high-quality historical data from Chinese social-media platforms. Ideally, the analysis would use archived Weibo hot-search data, Douyin trend data, WeChat article circulation, Xiaohongshu engagement, and moderation records. In practice, those data are difficult to obtain consistently across cases, especially after the events have already passed. This limitation shapes the paper's empirical strategy.

For that reason, this paper does not claim that Google Trends directly measures domestic Chinese social-media attention. This is especially important because Google is not normally accessible to ordinary users in mainland China, and Google Trends measures normalized search interest rather than social-media discussion volume [1]–[3]. Treating Google Trends as a direct substitute for Weibo, Douyin, WeChat, or Xiaohongshu would therefore be methodologically misleading.

Instead, Google Trends is used as a secondary proxy for broader cross-platform public salience. The logic is that when a China-based controversy becomes large enough, it may travel beyond its original domestic platform environment and appear in overseas Chinese discussion, Chinese- and English-language news coverage, investor search behavior, general public search interest, and interpersonal discussion. Google Trends is useful for identifying these externalized attention peaks, but it cannot reveal the full volume, sentiment, mod-

TABLE I
CASE SELECTION LOGIC

Case	Trigger	Main pathway	Primary spillover	Why included
Nongfu Spring	Zong Qinghou's death and brand controversy	Brand boycott	Reputational damage and stock decline	Consumer-goods brand spillover
Salt panic	Japan wastewater release	Panic buying	Hoarding, salt-stock volatility, reassurance	Fast commodity panic
Lianhua Qingwen	Zero COVID relaxation and medicine concern	Medicine demand	Product attention and Yiling repricing	Product-linked demand shock
Urumqi fire	Residential fire and protest visibility	Boundary policy pathway	Government response and reopening repricing	Politically sensitive boundary case

TABLE II
GOOGLE TRENDS VARIABLES

Case	Google Trends variables
Nongfu Spring	"Nongfu Spring," "Wahaha"
Salt panic	"Japan nuclear wastewater," "salt panic"
Lianhua Qingwen	"Lianhua Qingwen," "Yiling Pharmaceutical," "China COVID medicine"
Urumqi fire	"China protests," "zero covid," "Urumqi fire"

eration structure, or algorithmic pathway of domestic Chinese social-media discussion.

The paper therefore triangulates Google Trends with other evidence: event timelines, media reports, stock-price and trading-volume data, official statements, policy actions, and survey results. In this design, Google Trends is not the main proof of Chinese public opinion. It is one imperfect but comparable signal of whether a controversy had moved into a broader public-attention environment.

Unless otherwise noted, case figures combine Google Trends search-interest data with Investing.com historical-price and trading-volume data. Event windows are reconstructed from cited media and official sources. Survey visualizations are based on the author survey output files in `survey_outputs/` [3], [19].

V. SURVEY-BASED TRIANGULATION OF EVENT AWARENESS AND PERCEIVED PLATFORM CONTROL

To reduce reliance on Google Trends alone, the paper adds survey evidence from 104 respondents. The sample is not nationally representative. It is best understood as a convenience sample of students, young professionals, Chinese immigrants, and international students who have experience with Chinese social media and some familiarity with U.S. or Western social media. This sample is useful for studying perceived cross-platform awareness, but it cannot estimate national-level public opinion in China.

The survey supports the paper's decision to treat the selected cases as recognizable high-visibility events. Most respondents had heard of the four events before taking the survey. Recognition was above 60 percent for the Urumqi fire, the salt panic related to Japan's wastewater release, the Nongfu Spring controversy, and the Lianhua Qingwen debate. This matters because the paper's cases are not obscure examples

TABLE III
KEY SURVEY FINDINGS

Survey item	Main result
Heard of Urumqi fire	71.2%
Heard of Nongfu Spring controversy	67.3%
Heard of salt panic	63.5%
Heard of Lianhua Qingwen debate	63.5%
First learned through Chinese social media	38.5%
Agree algorithms/trending lists increased visibility	60.6% rated 4–5
Agree moderation/censorship reduced visibility	61.6% rated 4–5
Agree government likely responds to large online trend	85.6% rated 4–5

selected after the fact. They were broadly recognizable among respondents with relevant media experience.

Table III summarizes the key survey findings. Appendix Fig. A1 reports the cleaned survey visualizations.

The survey also supports the paper's cross-platform framing. A substantial share of respondents first learned about these events through Chinese social-media platforms such as Weibo, Douyin, WeChat, or Xiaohongshu. Others first encountered them through traditional news media, Western social media, or friends, family, and classmates. This pattern suggests that the events did not remain confined to one platform. They moved through a mixed attention environment in which social media, news, interpersonal networks, and search behavior interacted.

The results further show that respondents perceived platform control as both amplifying and limiting visibility. Many respondents agreed that platform algorithms or trending lists increased the visibility of these events. At the same time, many also agreed that moderation, censorship, or platform intervention reduced visibility. This is important for the paper's argument because Chinese platform control should not be understood as only suppression or only amplification. Visibility is produced through both mechanisms. Some topics are promoted, some are restricted, and some become consequential precisely because they break through this controlled environment.

The survey also connects online visibility to perceived offline effects. Most respondents agreed that online trends can create real changes in sales, shortages, or brand reputation. Many respondents also reported that online discussion affected their own consumption behavior, including looking up more

information before buying, delaying or avoiding a purchase, switching brands or products, or buying extra products. These responses do not prove that online trends caused the market movements observed in the case studies, but they support the plausibility of the consumer-spillover mechanism.

Finally, the survey supports the boundary-case logic. Most respondents agreed that when an online trend becomes large enough in China, the government is likely to respond publicly. Many also identified likely responses such as official clarification, regulatory or administrative action, market intervention, or supply reassurance. This does not prove that government action is caused by online attention, but it helps explain why the Urumqi fire and salt panic are relevant to a framework about platform control and offline response.

Overall, the survey is used as perception-based triangulation. It does not replace platform data, and it does not establish causality. Its value is to show that the selected cases were recognizable, that respondents associated them with Chinese social media and cross-platform visibility, and that respondents perceived both consumer and government spillovers as plausible outcomes of large online trends.

VI. RESULTS

A. Results Overview

The case analysis proceeds from consumer-goods spillovers to the boundary policy case. The first three cases examine different ways that social-media-originated attention can become associated with consumer behavior and market response. The fourth case examines a politically sensitive event in which visibility was associated with government response rather than consumer-goods behavior.

The comparison is organized around three questions. First, what triggered the online controversy? Second, how did the controversy move from platform-specific discussion into broader public attention? Third, what type of spillover followed: brand damage, panic buying, product-demand repricing, or government response?

B. Case 1: Nongfu Spring and Brand-Boycott Spillover

The Nongfu Spring case represents a brand-boycott pathway. The trigger was not a formal regulatory event or a public-health emergency, but a reputational controversy that developed after the death of Wahaha founder Zong Qinghou in February 2024. Online sympathy for Wahaha became linked to criticism of Nongfu Spring, and the discussion developed into a broader boycott narrative around brand identity, nationalism, and consumer loyalty [20]. Figure 1 links the attention breakout to the beginning of the stock-market decline.

The key point in this case is that the controversy did not matter only as online discussion. It became consequential when the brand dispute moved into broader public salience. Search interest, media discussion, investor attention, and stock-market movement became temporally aligned, as shown in Figure 1. In this sense, the case fits the paper's broader argument: a controversy becomes more important when it breaks out of platform-specific discussion and becomes legible

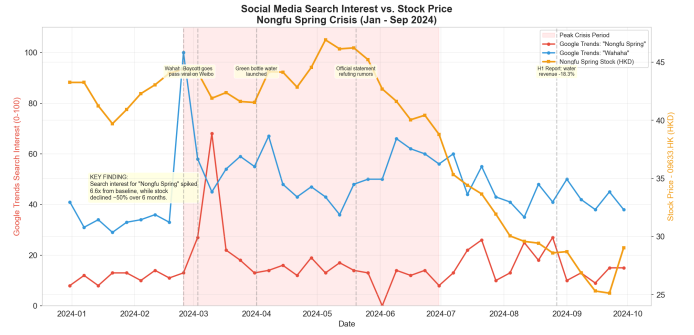


Fig. 1. Case 1: weekly search attention and Nongfu Spring stock price. The figure highlights the early-March visibility break and helps distinguish threshold crossing from the longer-run persistence that follows.

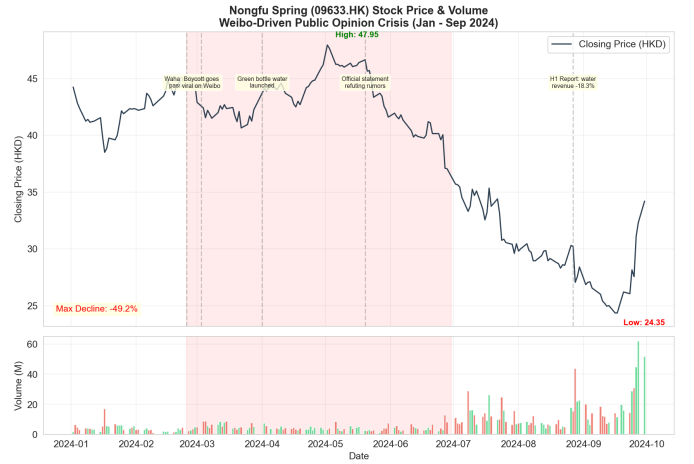


Fig. 2. Case 1: Nongfu Spring price and volume. The persistent decline and elevated trading activity distinguish slow-burn reputational damage from a short correction cycle.

to consumers and investors. Figure 2 then shows that the price decline and heavier trading persisted beyond the initial trigger window.

The main spillover was persistent reputational and market damage rather than a short panic spike. The stock decline was not limited to the initial attention burst, which suggests that the controversy became incorporated into a longer reassessment of brand value and consumer sentiment. This makes Nongfu Spring different from the salt panic case, where the attention spike and market response were sharper but shorter.

The case does not prove that online attention alone caused the stock decline. Other business, market, and financial factors may also have contributed. However, the timing supports the interpretation that online visibility was associated with reputational spillover. Among the three consumer-goods cases, Nongfu Spring shows the slowest but most persistent pathway.

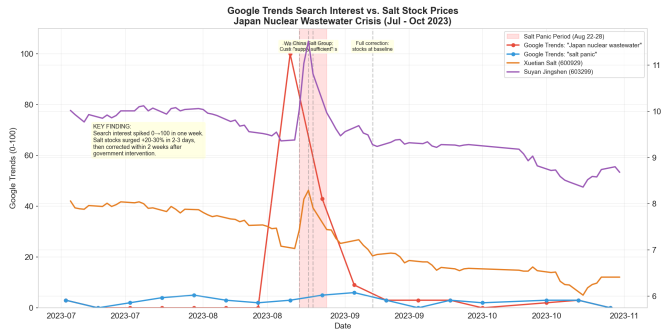


Fig. 3. Case 2: weekly salt-panic attention and salt-sector stocks. The figure shows how sharply the attention burst rises and falls, which supports the panic-coordination interpretation.

C. Case 2: Japan Wastewater Salt Panic and Panic-Buying Spillover

The salt panic case represents a panic-buying and supply-reassurance pathway. The trigger was Japan's release of treated nuclear wastewater in August 2023, which generated public concern about food safety and salt supply. Online discussion helped transform a geopolitical and public-health concern into a consumer-goods panic. The result was not only discussion, but hoarding behavior, salt-sector stock volatility, and official reassurance about supply [15], [21], [22]. Figure 3 shows how the attention surge and salt-sector stock reaction concentrate in the same short window.

This case is important because the spillover was immediate. Search attention rose sharply, consumer panic appeared quickly, and market and government responses occurred in the same narrow window, as shown in Figure 3. Compared with the Nongfu Spring case, the salt panic was shorter and more intense. It shows how online visibility can coordinate behavior around a simple consumer action: buying and storing salt. Figure 4 then shows the burst-period volume spike and the quick reversal that followed.

The market response was visible in both price movement and trading volume among salt-related firms. However, the more important analytical point is not only the stock movement. It is the combination of consumer hoarding and government supply reassurance. The government response suggests that when an online trend creates visible market disorder or shortage expectations, the state may respond not only by moderating discussion but also by reassuring supply, warning against hoarding, or stabilizing prices [22].

The salt panic therefore supports the paper's claim that consumer-goods spillovers can be both market-based and governance-based. It is still not a pure causal test, because the wastewater release itself was a major external event. But the timing suggests that online visibility amplified the consumer response and made the issue visible enough for state intervention.

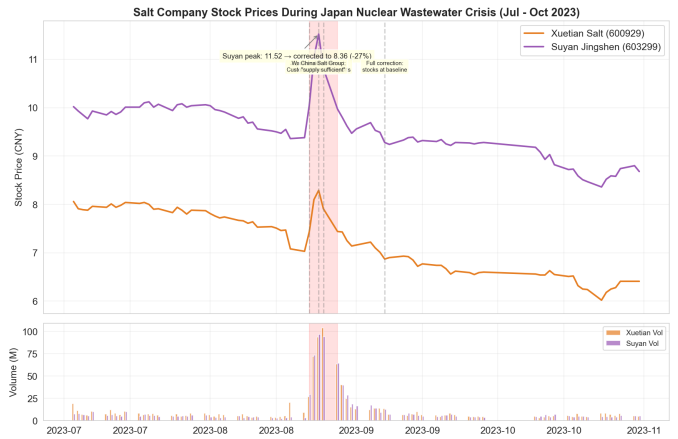


Fig. 4. Case 2: salt-stock price and volume. Extraordinary burst-period volume and the quick reversal help distinguish panic coordination from the longer persistence seen in Case 1.

D. Case 3: Lianhua Qingwen and Medicine-Demand Repricing

The Lianhua Qingwen case represents a medicine-demand and pharmaceutical-repricing pathway. Unlike Nongfu Spring, this was not a boycott. Unlike the salt panic, it was not a simple hoarding event around a basic commodity. The trigger was the reopening period after China began moving away from Zero COVID controls in late 2022. As infection concerns increased, public attention turned toward COVID-related medicines, including Lianhua Qingwen [23], [24].

This case shows how online attention can shape demand expectations. The controversy around Lianhua Qingwen was connected to both public-health uncertainty and consumer search for treatment options. Product-specific attention became embedded in a broader reopening-era health-information environment, and Figure 5 places that surge alongside Yiling Pharmaceutical's market movement. This distinction matters because the case should not be interpreted as a general pharmaceutical rally. The strongest market response was concentrated around the company most directly linked to the product [14], [15], [24].

The comparison with Tongrentang helps clarify the mechanism. If the effect were only a broad traditional-medicine or pharmaceutical-sector rally, both firms would be expected to move similarly. Instead, the product-linked firm showed a stronger rally and clearer correction. This pattern supports the interpretation that product-specific online attention and medicine-demand expectations were associated with the repricing. Figure 6 then compares the two stocks directly and shows the sharper rally-correction cycle in Yiling.

The Lianhua Qingwen case also shows that spillovers can reverse. Online attention may produce short-term demand expectations, but once the initial demand shock fades or expectations adjust, market correction can follow. This makes the case analytically different from Nongfu Spring's persistent

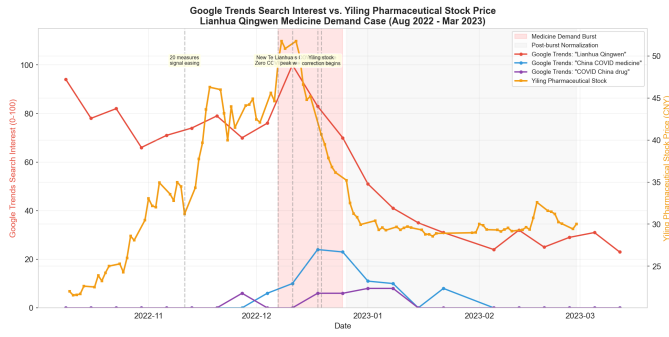


Fig. 5. Case 3: weekly Lianhua Qingwen attention and Yiling Pharmaceutical stock price. The figure highlights the post-reopening attention surge and its concentration around the product-linked firm.

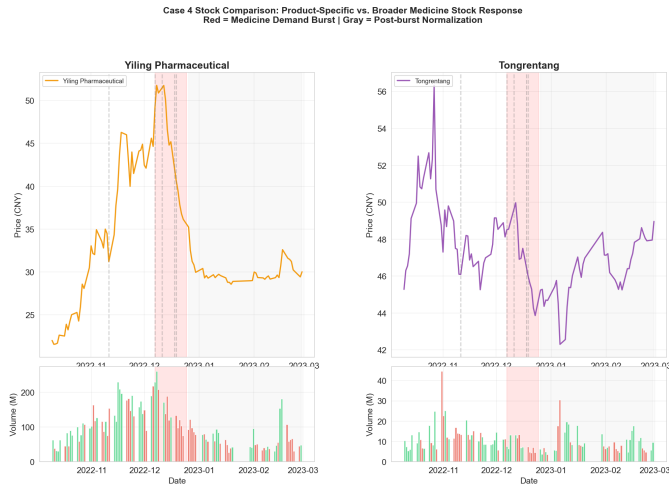


Fig. 6. Case 3: Yiling Pharmaceutical versus Tongrentang. The comparison shows stronger direct exposure and a clearer correction in Yiling, which helps distinguish product-linked repricing from a broad sector rally.

reputational damage and the salt panic's short commodity shock.

E. Boundary Case: Urumqi Fire, Protest Visibility, and Government Response

The Urumqi fire is treated as a boundary case rather than a consumer-goods case. Its inclusion is justified because it shows what happens when online attention becomes politically sensitive and difficult to contain. The trigger was the residential fire in Urumqi on November 24, 2022. Online mourning and anger became connected to broader criticism of Zero COVID controls, and the event became part of a wider protest signal [25]. Figure 7 situates the protest-attention spike alongside Air China's reopening-sensitive price movement.

The main spillover in this case was not consumer purchasing behavior. It was policy pressure. Figure 7 situates the protest-related attention window alongside reopening-sensitive market movement. The case therefore marks the outer boundary of the paper's argument: when online attention becomes sufficiently visible, platform control may shape the pathway of visibility, but it may not fully prevent the issue from becoming legible to the government.

The event sequence suggests a compressed relationship between protest visibility, policy change, and market repricing. Reopening-sensitive stocks began to move around the protest and policy-reversal window, while the December 7 "New Ten Measures" marked a major change in China's Zero COVID policy [23], [26]. This does not mean that online attention alone caused the policy shift. The policy environment was already under pressure from economic, epidemiological, and administrative factors. However, the case shows how a politically sensitive attention event can become associated with state response once it reaches sufficient visibility [17], [18]. Figure 8 then shows that the repricing extended beyond Air China to Haidilao and Trip.com.

As a boundary case, Urumqi helps clarify the difference between consumer-goods spillover and policy spillover. In the first three cases, online visibility is associated mainly with consumer behavior, brand perception, or market expectations. In the Urumqi case, visibility is associated with governance response. The case therefore supports the broader framework while remaining analytically distinct from the consumer-goods cases.

F. Cross-Case Comparison

The four cases show that visibility breakthroughs do not produce one uniform type of offline effect. Instead, the spillover depends on the event type.

Nongfu Spring shows a brand-boycott pathway. The controversy developed through reputational conflict and produced a longer-lasting market decline. Salt panic shows a panic-buying pathway. The trigger was sudden, the attention spike was intense, and the response included both consumer hoarding and official supply reassurance. Lianhua Qingwen shows a medicine-demand pathway. The attention burst was connected to reopening-era health uncertainty, producing product-linked repricing followed by correction. Urumqi shows a boundary policy pathway. The attention was politically sensitive, and the

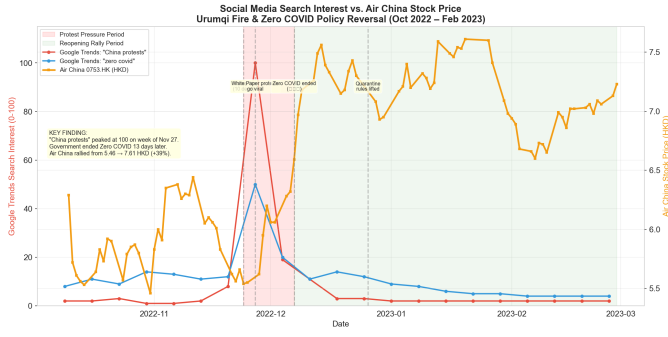


Fig. 7. Boundary case: protest-related search attention and Air China stock price. The figure compresses visibility, policy reversal, and reopening-sensitive repricing into one short window.

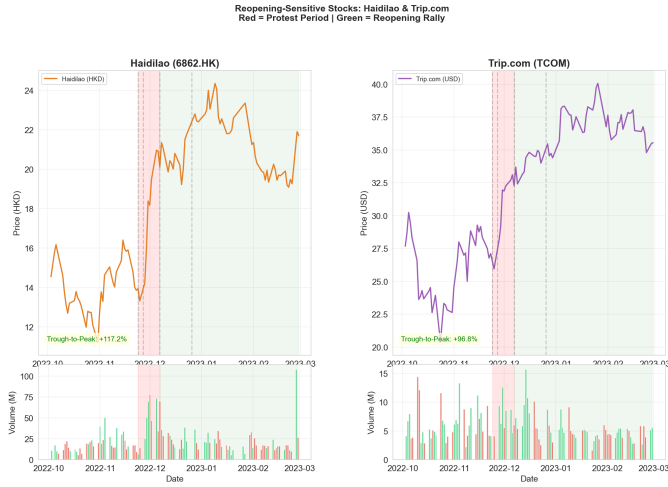


Fig. 8. Boundary case: Haidilao and Trip.com during the protest and reopening windows. The comparison shows that repricing spread across reopening-sensitive firms rather than staying confined to one stock.

main spillover was government response rather than consumer behavior.

This comparison supports the paper’s main claim. Online controversies in China become consequential not simply because they are discussed online, but because they move beyond their original platform spaces and become visible to actors outside the platform: consumers, investors, firms, regulators, or state agencies. The same visibility breakthrough can produce

different outcomes depending on whether the issue concerns a brand, a basic commodity, a health product, or a politically sensitive event.

At the same time, the comparison also limits the paper’s claims. The cases are not representative of all online controversies in China. They are most-likely cases selected because each has a visible trigger and observable spillover. The findings should therefore be read as a theory-building comparison rather than as a population-level test.

VII. DISCUSSION

The paper’s main finding is that online attention becomes consequential through visibility breakthrough, not through platform discussion alone. In China’s governed platform environment, visibility is shaped by both amplification and restriction. Algorithms, trending lists, influencer participation, reposting, and short-video circulation can increase visibility, while moderation, censorship, search filtering, and platform intervention can reduce or redirect it. The survey results suggest that respondents perceive both processes at the same time: Chinese platform control is not viewed as simply suppressive, but as a system that can both amplify and constrain public attention.

This helps explain why the paper uses a threshold framework. A controversy may exist online without producing obvious offline consequences. But when it becomes visible enough to move across platforms and into broader public attention, it can become legible to consumers, investors, firms, and government actors. In the cases examined here, that visibility breakthrough is associated with different forms of spillover: brand damage, panic buying, medicine-demand repricing, and policy response.

The Google Trends limitation is central to this interpretation. Since Google Trends cannot directly measure domestic Chinese social-media discussion, it should not be overread. Its value is narrower: it helps identify moments when China-related controversies became visible in a broader search and public-attention environment. The survey evidence partially strengthens this interpretation by showing that the selected events were widely recognized among respondents with Chinese social-media experience and that many respondents perceived both consumer and government effects from large online trends.

The boundary case also matters. The Urumqi fire shows that platform control does not mean full control over public

TABLE IV
CROSS-CASE COMPARISON

Case	Trigger	Pathway	Spillover	Duration	Main inference
Nongfu Spring	Brand controversy	Brand boycott	Reputation and stock decline	Persistent	Longer market reassessment
Salt panic	Wastewater release	Panic buying	Hoarding, volatility, re-assurance	Short and intense	Immediate market and governance response
Lianhua Qingwen	Medicine concern	Product demand	Repricing and correction	Burst/correction	Product-linked demand expectations
Urumqi fire	Fire and protest visibility	Boundary policy pathway	Government response and reopening repricing	Compressed	Policy response, not consumer-goods spillover

attention. Under certain conditions, politically sensitive attention may become visible enough that policy response becomes more likely. This does not prove that online attention caused the Zero COVID reversal, but it shows that major online visibility can become part of a broader pressure environment.

VIII. LIMITATIONS

This paper has several limitations.

First, the case selection is purposive rather than representative. The four cases were selected because they are recent, visible, and data-accessible. They should not be treated as a random sample of Chinese online controversies.

Second, the paper does not have direct historical platform data from Weibo, Douyin, WeChat, or Xiaohongshu. This prevents the analysis from measuring the full domestic discussion volume, sentiment structure, moderation process, or algorithmic pathway of each event.

Third, Google Trends is an imperfect proxy. It is useful for observing broader cross-platform public salience, but it cannot directly represent Chinese domestic social-media attention. The paper therefore uses Google Trends only as one signal among several.

Fourth, the survey is not nationally representative. The respondents are mainly students, young professionals, Chinese immigrants, and international students with Chinese social-media experience and some familiarity with U.S. or Western social media. The survey is useful for perception-based triangulation, but not for estimating Chinese public opinion as a whole.

Fifth, the paper cannot establish causality. Market movements, consumer behavior, and government response may be shaped by broader news shocks, firm fundamentals, policy changes, economic conditions, and public-health developments. The paper therefore uses cautious language such as “associated with,” “temporally aligned with,” and “suggests,” rather than claiming direct causal effects.

IX. CONCLUSION

This paper examined how social-media-originated controversies in China move beyond platform-specific discussion and become associated with offline spillovers. Its central argument is that online trends become consequential when they break through a governed visibility environment and enter broader public attention. Once this happens, the spillover pathway depends on the type of event.

The three consumer-goods cases show different mechanisms. Nongfu Spring illustrates how online brand controversy can become associated with reputational damage and market decline. The salt panic shows how public-health fear can become associated with panic buying, commodity-stock volatility, and government supply reassurance. Lianhua Qingwen shows how reopening-era health uncertainty can become associated with product-specific medicine demand and pharmaceutical repricing. The Urumqi fire, treated as a boundary case, shows how politically sensitive attention can become associated with government response and policy change.

The paper does not claim that Google Trends directly measures Chinese social-media attention. Nor does it claim that online attention alone caused the observed market or policy outcomes. Instead, it uses Google Trends as a secondary indicator of broader public salience and triangulates it with survey responses, event timelines, market indicators, media reports, and policy actions. This approach is limited, but it gives a more careful way to study online spillover when direct platform data are unavailable.

The broader implication is that platform control does not eliminate online spillover. It changes the conditions under which spillover occurs. In China, some online controversies may remain contained, fragmented, or short-lived. Others break through the visibility threshold and become meaningful to consumers, firms, investors, or the government. Understanding that threshold is essential for explaining when online attention becomes more than online attention.

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APPENDIX A
SUPPLEMENTARY FIGURES

Figure A1 retains one cleaned survey summary visualization from `survey_outputs/`.

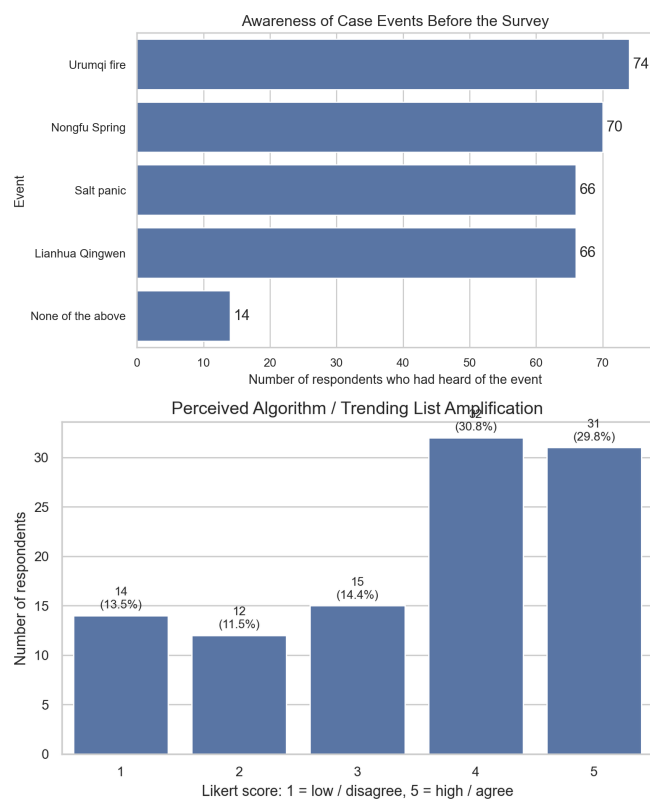


Fig. A1. Survey-based triangulation of event awareness and perceived platform control. Data: author survey, $n = 104$.